

Highly selective striatal response to visual feedback in reward-based motor skill learning

HANYANG UNIVERSITY
SEOUL, KOREA

Sungbeen Park¹, Junghyun Kim², Sungshin Kim^{1,2,3*}

¹Department of Artificial Intelligence, Hanyang University, Seoul, South Korea

²Department of Intelligence Computing, Hanyang University, Seoul, South Korea

³Center for Neuroscience Imaging Research, Institute for Basic Science, Suwon, South Korea



Background

- The striatum, a key region of the basal ganglia, is frequently involved in both reward processing and motor control. During motor learning, it is typically difficult to distinguish between these two functions.
- To untangle striatum's roles, we designed an fMRI experiment of learning a completely novel motor task guided by different conditions of visual feedback.
- Highly selective responses in the striatum related to task performance support the idea that the visual feedback plays a role of a reward signal, potentially reflecting the dopamine release.
- Additionally, we further examined dissociable roles of striatal subregions in reward processing and motor control.
- Finally, we discussed the respective role of the cortical and subcortical regions in model-based versus model-free reinforcement learning, depending on online visual feedback versus binary visual feedback.

Experimental design

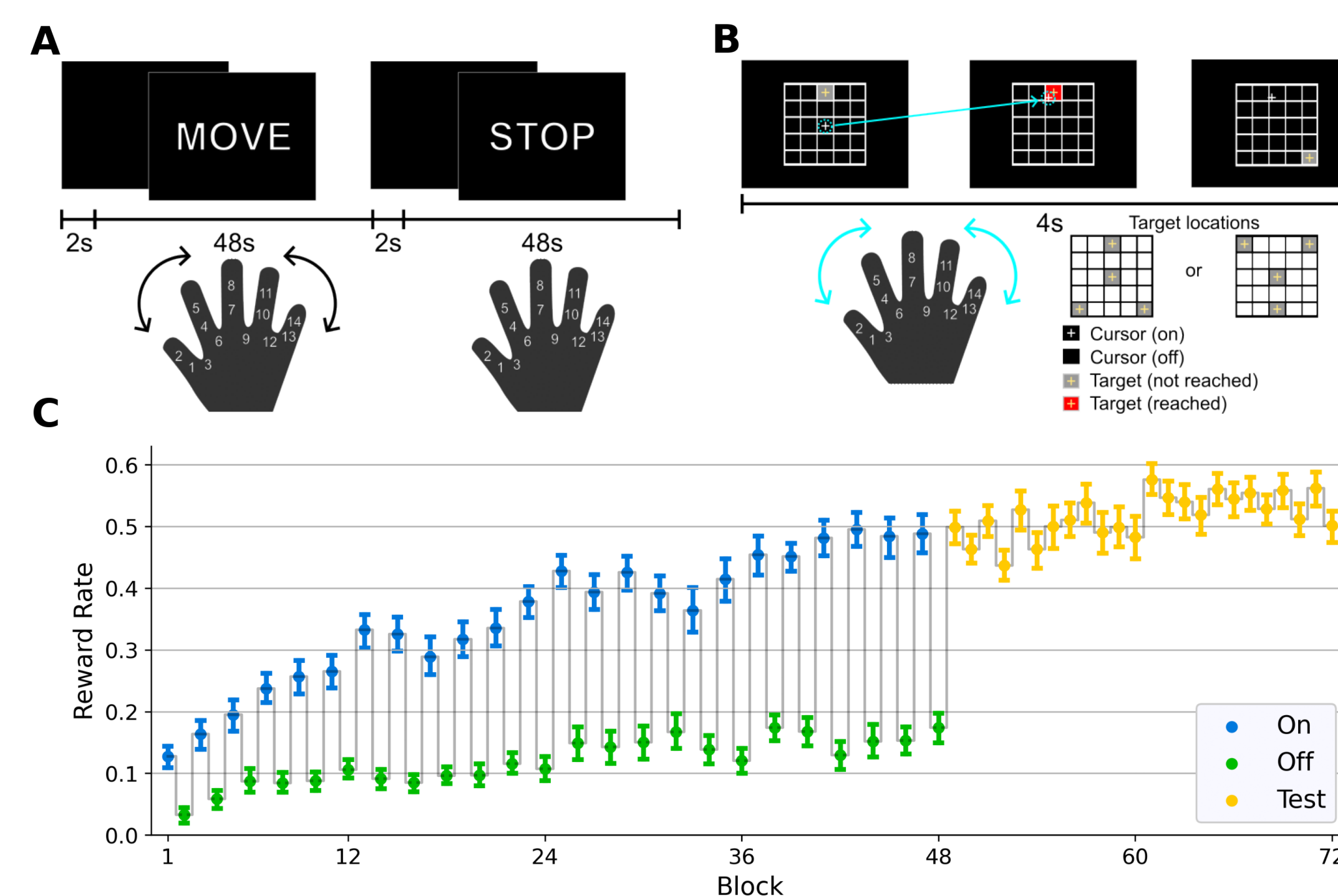


Figure 1. Experimental design and behavioral performance of all participants.

For about an hour, the 24 participants used a data glove to perform an fMRI experiment.

- (A) Participants should **move** or **stop** their right fingers freely in response to the message displayed on the screen during the **localizer session**.
- (B) During the **main session**, targets presented in the same sequence for each block, and participants performed the same task when the **cursor** was visible in odd blocks but not in even blocks.
- (C) Block-by-block group performance. **Visual feedback (blue)** improves performance compared to **no feedback (green)**, and learning continues during the test session (yellow).

References

- Jörn Diedrichsen and Katja Kornysheva. Motor skill learning between selection and execution. *Trends in cognitive sciences*, 19(4):227–233, 2015.
- Julien Doyon, Virginia Penhune, and Leslie G Ungerleider. Distinct contribution of the cortico-striatal and cortico-cerebellar systems to motor skill learning. *Neuropsychologia*, 41(3):252–262, 2003.
- Yi Huang, Zachary A Yaple, and Rongjun Yu. Goal-oriented and habitual decisions: Neural signatures of model-based and model-free learning. *NeuroImage*, 215:116834, 2020.
- Matthias J Koepp, Roger N Gunn, Andrew D Lawrence, Vincent J Cunningham, Alain Dagher, Tasmin Jones, David J Brooks, Christopher J Bench, and PM Grasby. Evidence for striatal dopamine release during a video game. *Nature*, 393(6682):266–268, 1998.
- Stéphane Lehéricy, Habib Benali, Pierre-François Van de Moortele, Mélanie Péligrini-Issac, Tobias Waechter, Kamil Ugurbil, and Julien Doyon. Distinct basal ganglia territories are engaged in early and advanced motor sequence learning. *Proceedings of the National Academy of Sciences*, 102(35):12566–12571, 2005.

Whole brain voxel-wise GLM analysis

- We hypothesized the red color, to which a target changes when virtually reached, would serve as a **reward signal**.
- To identify brain regions related to reward processing, we calculated the task performance as a proportion of time turning on red for each trial, every 4 seconds.

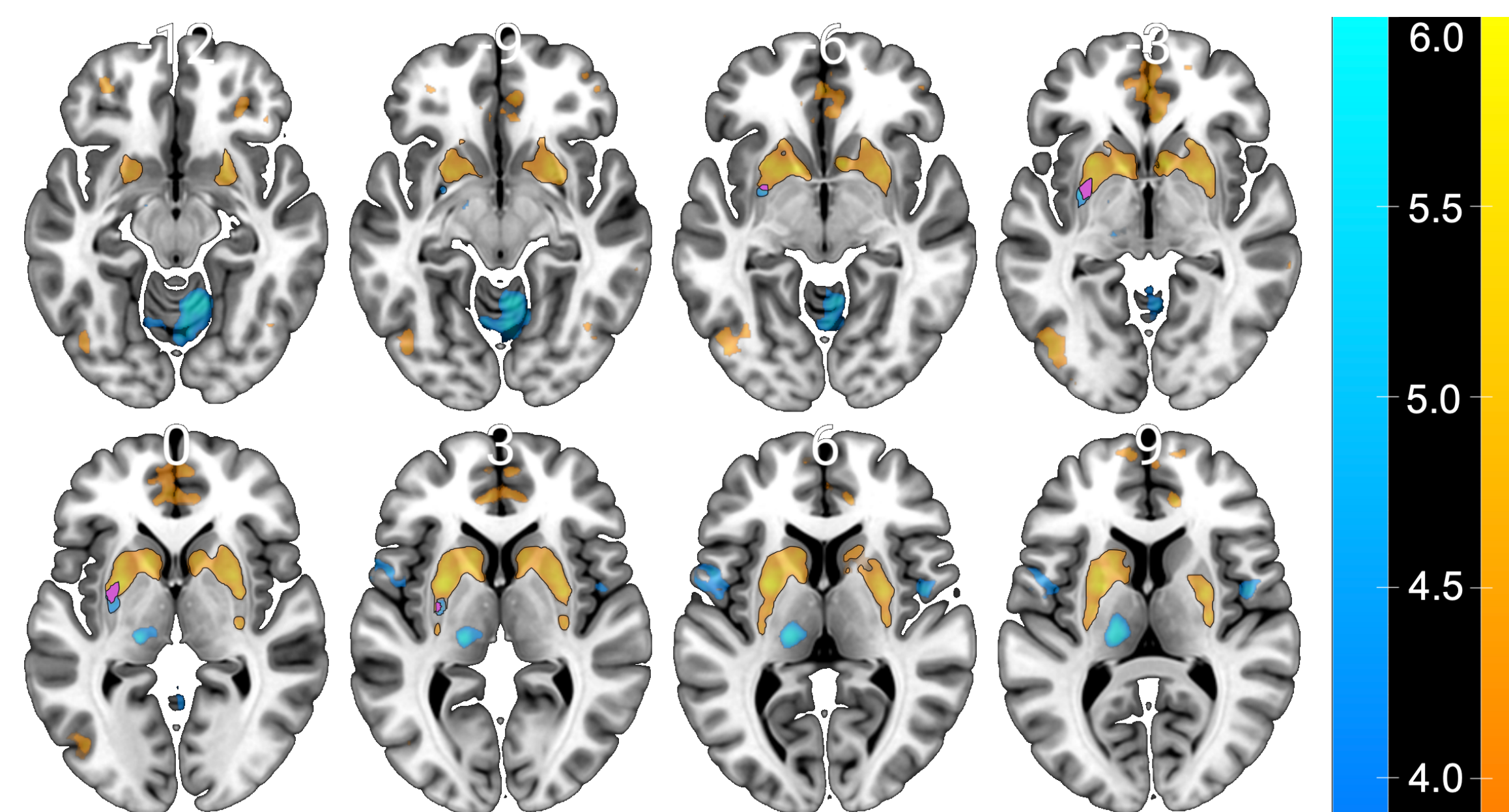


Figure 2. Brain regions positively related to the performance (yellow), movement (blue), and both (purple).

- Are the striatal subregions **functionally dissociated** for reward processing and motor control?
- Do the cortical (vmPFC) and subcortical (striatum) regions play distinct roles in **model-based** versus **model-free** reinforcement learning?

Functional delineation of the striatum

Only in the striatum did we find clusters larger than 80 voxels at $p < 10^{-5}$. This led us to analyze the GLM's sensitivity in defining the striatum.

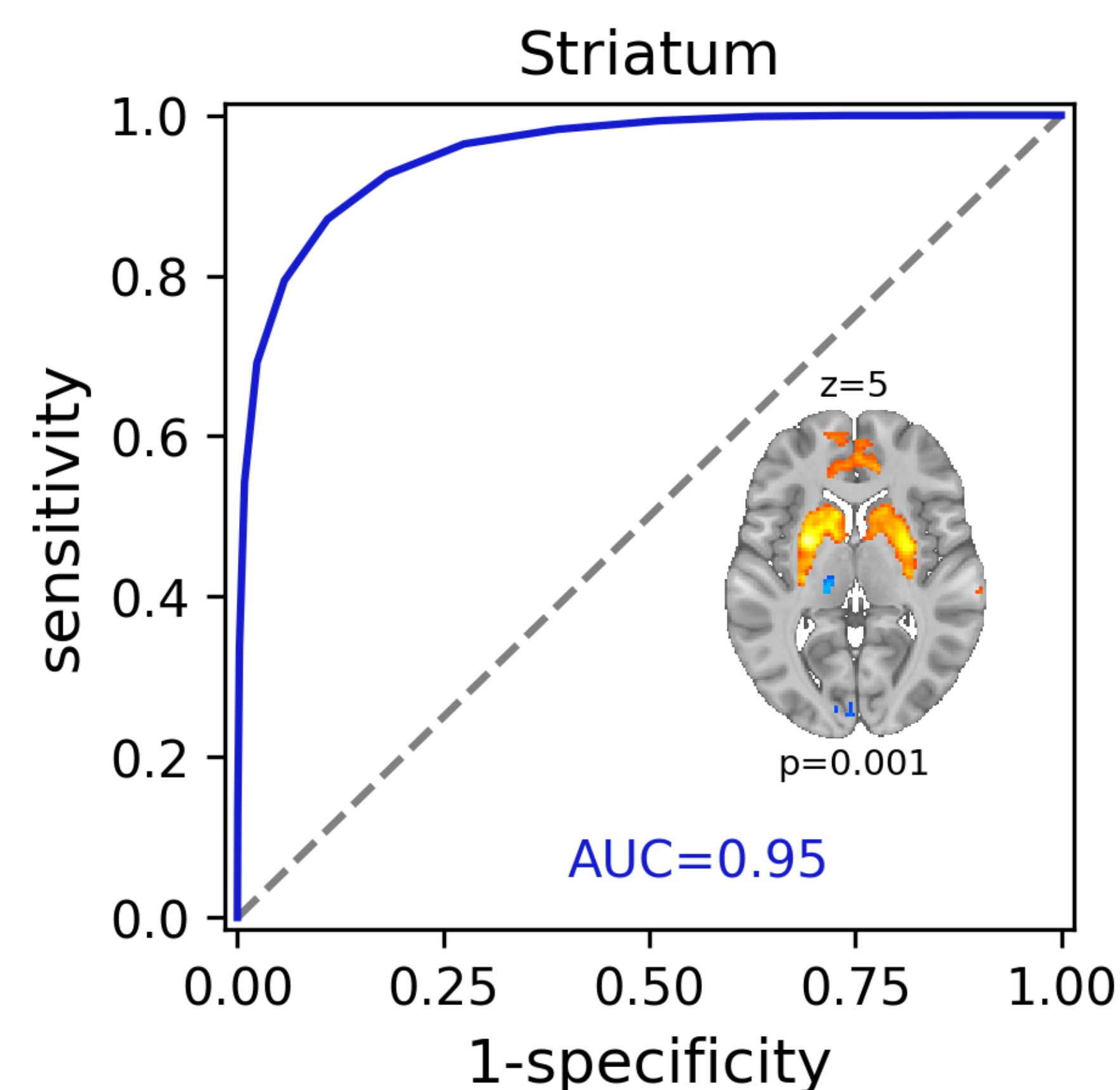


Figure 3. The receiver operating characteristic (ROC) curve (blue) for delineating an anatomically defined striatum.

Region of interest (ROI) analysis

Given the GLM results, we further conducted ROI analysis to investigate the dissociable activities of **striatal subregions** and the **ventromedial prefrontal cortex**.

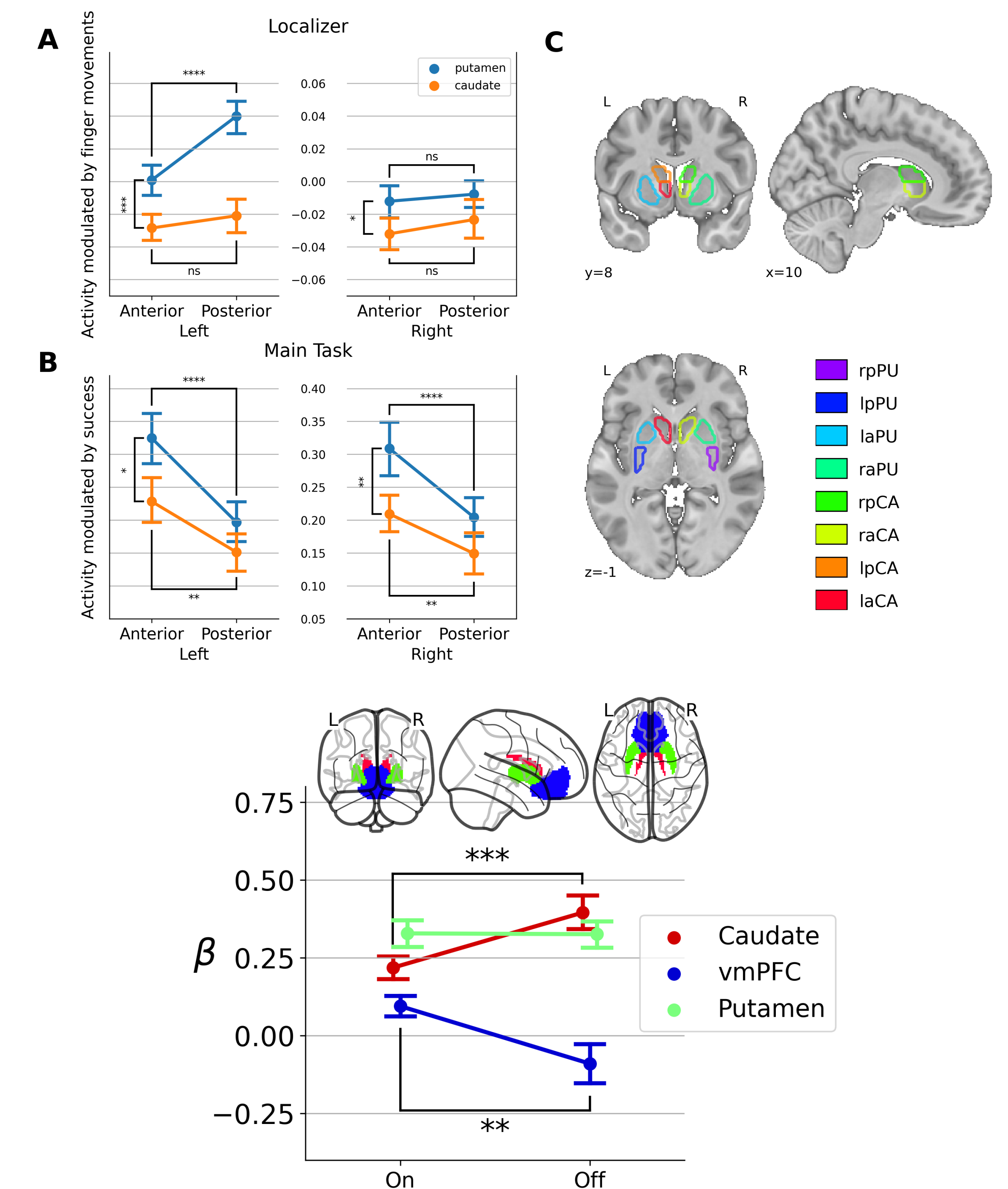


Figure 4. Spatial dissociation of fMRI activities in ROIs. (Top) the brain activity that success rate influences. (Bottom) Brain activity through a block design that toggles feedback.

Conclusion

- Our GLM analysis with a parametric regressor modulating the performance **delineated** the striatum with strikingly high sensitivity and specificity.
- The effects of **visual feedback** on the striatal response were spatially dissociated from those of motor control, respectively mapped on the **anterior striatum** and **contralateral posterior putamen**.
- It would be an interesting future work, **vmPFC** and **striatum**, found to be related to reward processing in the current study respectively contribute to **model-based** and **model-free** motor learning

Acknowledgments

This work was supported by the National Research Foundation of Korea (NRF-2021R1A2C2011648), Hanyang University (HY-20200000002753), and Center for Neuroscience Imaging Research, Institute for Basic Science, Korea (IBS-R015-D1)

Contact information: sungshinkim@hanyang.ac.kr